Two estimators, not a benchmark: reconciling DNA foundation models and MPRAs for regulatory variant effects in the developing human cortex

Author Name

Affiliation

Abstract. DNA foundation models and massively parallel reporter assays (MPRAs) are increasingly cross-compared to benchmark *in silico* variant-effect prediction, almost always treating the assay as ground truth. We argue this framing is mis-specified: a foundation model predicts a variant's effect in its *native* genomic locus, whereas an MPRA measures a short element removed from that locus, so the two estimate different quantities and need not agree. Benchmarking AlphaGenome and Borzoi against a developing-human-cortex lentiMPRA of 3,976 biallelic psychiatric-risk regulatory variants (76 differentially active), we find model and reporter effect sizes are uncorrelated (Spearman $\rho = -0.02$ overall; -0.13 on the 76 differentially active variants), and a from-scratch Bayesian measurement-error model shows this is not reporter noise (denoised $\rho \approx 0$). The sequence models instead agree strongly with one another ($\rho = +0.55$) and weakly with base-level motif disruption and conservation, forming a sequence-level cluster the reporter is absent from; a third independent model, Enformer, reproduces both the inter-model agreement and the null agreement with the reporter ($\rho = -0.01$, n.s.). To break the tie *without* privileging either instrument, we bring in an independent, developmentally matched, fine-mapped fetal-cortex eQTL atlas: on 370 overlapping variants in eQTL credible sets, the models' predicted effects track the endogenous eQTL effect (AlphaGenome $\rho = 0.24$, 95% CI [0.14, 0.33]; Borzoi $\rho = 0.28$) while the reporter does not ($\rho = -0.03$, n.s.), and the models' direction-of-effect concordance rises from chance to 80% as variants become confidently causal (PIP ≥ 0.5; 20/25, $p = 0.002$). We present this as corroborating, not decisive, evidence — it rests on 370 variants (25 at PIP ≥ 0.5). Asking further whether the reporter's residual potential propagates to endogenous expression when context is favorable — an Activity–Context–Propagation model fit from scratch — we find it does not at this sample size: neither the reporter nor a reporter×context interaction predicts held-out endogenous effect, intrinsic scatter dominates, and the largest class of reporter-active variants is endogenously silent but not explained by the measured context. A from-scratch latent-variable model that separates each source's noise from shared signal estimates the calibrated model↔reporter agreement at $\rho = -0.018$ (95% HDI [-0.064, $+0.027$]). We deliver a calibrated concordance map of all 3,976 variants rather than a per-variant verdict, and argue that a fine-mapped, developmentally matched endogenous QTL — not a decontextualized reporter — is the appropriate arbiter of *in silico* regulatory variant-effect prediction. Finally, we turn the map to use: within its concordant core — the 25 concordant, differentially active variants — we resolve the direction of allelic effect across reporter, both models, and the eQTL, finding that magnitude concordance predicts sign agreement only weakly (10 of 25), and we rank the nominated genes for druggability and disease provenance. Every nominated gene is a novel, undrugged target; *STIP1* (bipolar disorder) emerges as the strongest overall druggable candidate and *MFGE8* as the strongest by causal confidence, while variant-to-indication assignment, not target tractability, is the rate-limiting step toward a therapeutic hypothesis.

1. Introduction

A central promise of sequence-to-function deep learning is that the regulatory consequence of a genetic variant can be read directly from DNA sequence. Models such as AlphaGenome, Borzoi and Enformer, trained on large compendia of functional-genomics tracks (DNase, ATAC, ChIP, CAGE, RNA-seq) measured on the native genome in real cells, predict base-resolution regulatory activity and score a variant's effect *in silico* as the difference between alternate- and reference-allele predictions [2,3,4]. In parallel, massively parallel reporter assays (MPRAs) measure the regulatory activity of thousands of short sequences experimentally, and by testing reference against alternate alleles they quantify single-base variant effects at scale [1,5]. The two approaches are increasingly cross-compared, and the assay is usually taken as the benchmark against which the model is scored.

We take a different stance. A foundation model and an MPRA are both *imperfect estimators* of an underlying variant effect, each with its own systematic biases and measurement noise; neither is ground truth. This framing matters because the two instruments measure subtly different estimands (Figure 1). A foundation model predicts the activity of an element in its *native genomic locus*, with its native distal enhancers, three-dimensional contacts, and local chromatin state. A lentiMPRA measures the activity of a short (~200 bp) oligo removed from its native neighborhood and integrated at semi-random sites across the genome. Critically, the lentiMPRA reporter is *not episomal*: it is delivered by lentivirus, stably integrated into the host-cell genome, and chromatinized like any integrated provirus [1]. Both instruments therefore involve chromatin; what differs is *native-locus* versus *decontextualized ectopic-locus* context. A near-zero agreement between them is thus compatible with context-dependent variants, not prima facie evidence that either instrument is broken.

The consequence is methodological: if two instruments estimate different quantities, neither can adjudicate the other, and a benchmark that treats one as truth is mis-specified. We therefore (i) quantify model↔reporter agreement over 3,976 variants and rule out reporter measurement noise using a from-scratch hierarchical Bayesian model; (ii) place the sequence models in a multi-estimator agreement structure alongside base-level conservation and motif disruption; (iii) introduce an *independent*, developmentally matched, fine-mapped fetal-cortex eQTL atlas as an external anchor that privileges neither instrument; (iv) fit a from-scratch latent-variable generative model that puts a credible interval on the calibrated agreement after each source's own noise is removed, and validate that estimator by simulation-based calibration; and (v) recast the deliverable from a per-variant verdict into a calibrated concordance map.

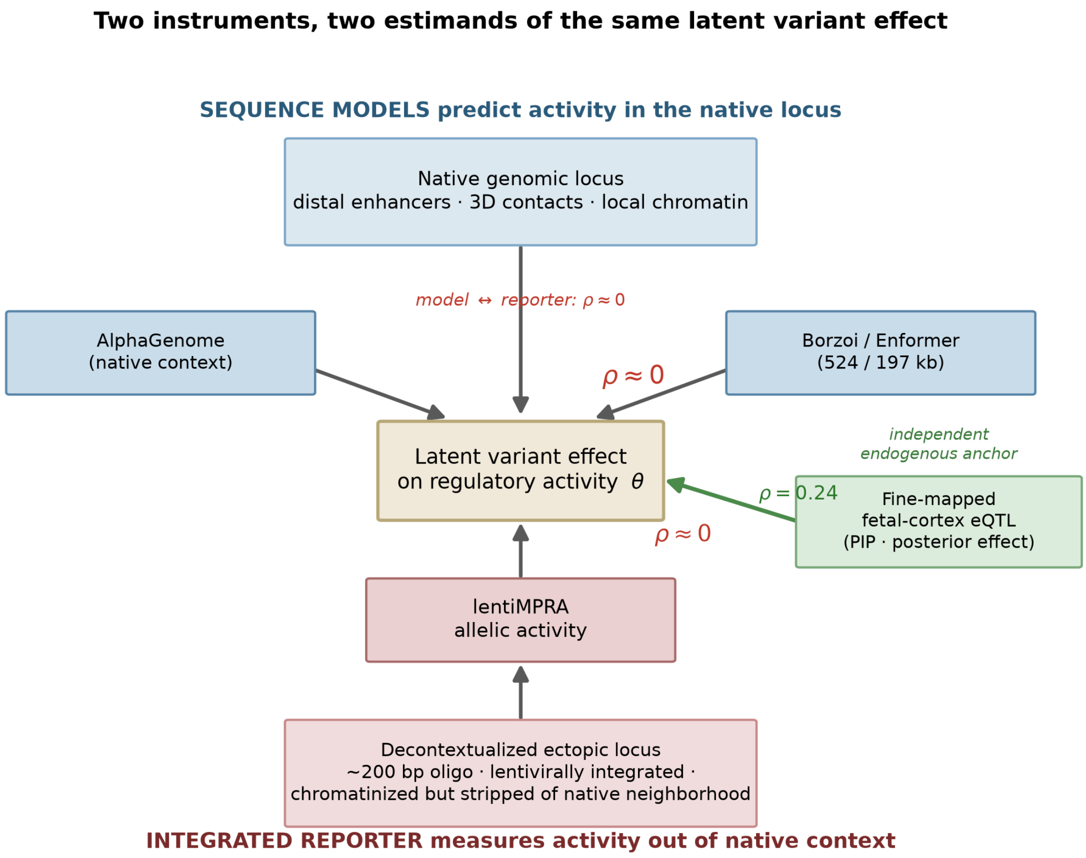

Figure 1. Two instruments, two estimands. A DNA foundation model predicts a variant's regulatory effect in its native genomic locus (top); an integrated lentiMPRA measures a decontextualized ~200 bp element chromatinized at ectopic integration sites (bottom). Both are imperfect readouts of a shared latent variant effect θ, but of different estimands, so near-zero model↔reporter agreement (ρ ≈ 0) is expected for context-dependent variants. A fine-mapped fetal-cortex eQTL (right) is an independent endogenous anchor that privileges neither instrument.

2. Results

2.1 Foundation-model and reporter variant effects do not agree

We scored all 3,976 biallelic variants with AlphaGenome in native genomic context and summarized each variant's predicted effect as the mean allelic difference across brain/CNS tracks (Methods). Compared against the lentiMPRA measured allelic effect ($\log_2$ fold change of alternate versus reference allele), the two estimators show no rank agreement. On signed effects the full-set Spearman correlation is −0.02 (Pearson −0.013), sign concordance 0.50 (chance); on effect *magnitude* it is $\rho = -0.01$ ($p = 0.43$; Figure 2a). The agreement is $\rho = -0.13$ on the 76 differentially active variants (DAVs), and predicted effect magnitude discriminates DAVs at an AUROC of only 0.54. The absence of correlation is not an artifact of allele orientation or genome build — a reference-allele check against hg38 passed on every sampled variant — nor of the particular track set or aggregation, which we varied without effect ($\rho \geq 0.98$ across choices).

That the model is nonetheless producing meaningful predictions is established by a positive control: AlphaGenome's predicted effect magnitude correlates with motifbreakR motif-disruption magnitude, a fully independent base-level sequence estimator ($\rho = 0.10$, $p < 0.01$, $n = 702$), whereas the reporter correlates with neither motifbreakR ($\rho = -0.02$, n.s.) nor the model (Figure 2b). Two sequence-based estimators agree with one another; the integrated reporter agrees with neither.

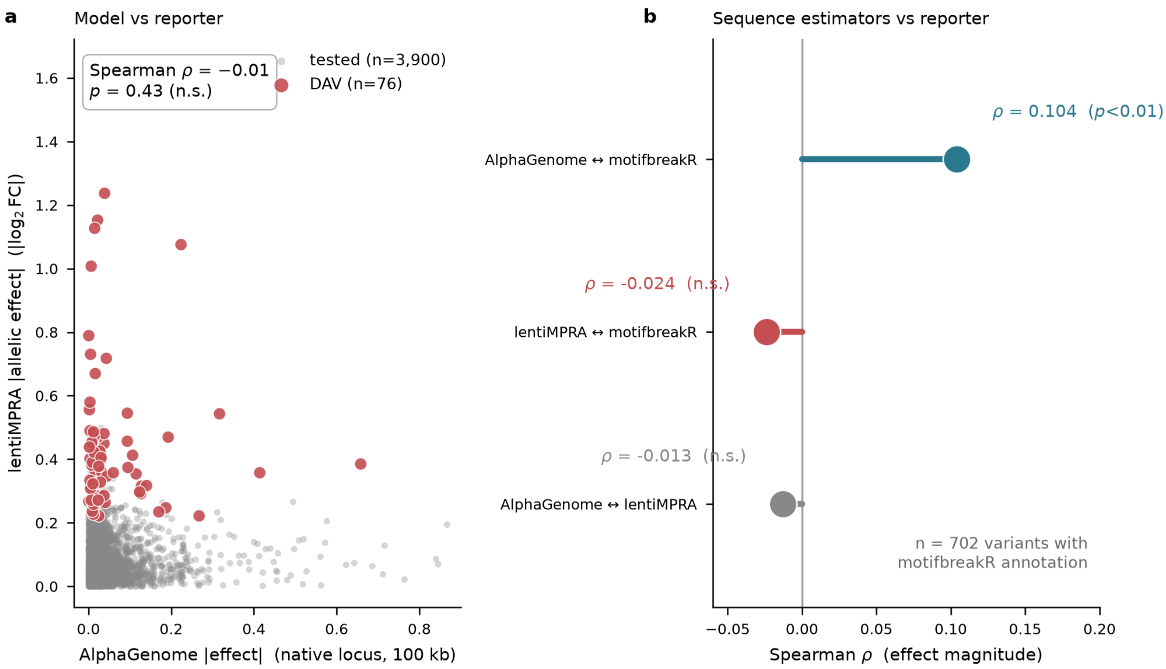

Figure 2. Foundation-model and integrated-reporter variant effects do not agree. (a) AlphaGenome predicted effect magnitude (native locus, 100 kb context) versus lentiMPRA measured allelic effect magnitude for all 3,976 variants; the 76 DAVs are highlighted. The cloud is structureless (Spearman $\rho = -0.01$, n.s.). (b) Rank agreement of effect magnitude for three estimator pairs: two independent sequence estimators (AlphaGenome, motifbreakR) agree ($\rho = 0.10$, $p < 0.01$), while the integrated reporter correlates with neither ($n = 702$).

2.2 The disagreement is not reporter measurement noise

A skeptic's first explanation for zero agreement is that the reporter measurements are too noisy. We tested this with a from-scratch hierarchical Bayesian measurement-error model (NumPyro): each variant's measured effect is a noisy draw around a latent true effect, using the assay's own limma-moderated standard error as the known measurement noise, with a regularized horseshoe prior encoding that most variants are null and a few have large effects (Methods). The fit was clean (maximum $\hat{R} = 1.007$, no divergences) and independently recovered the assay's own significance calls — the 95% credible interval excludes zero for 73 of the 76 limma DAVs without ever using the limma calls.

Denoising does not rescue the agreement. Recomputing model↔assay agreement on the posterior (denoised) effects, and then reliability-weighting by inverse posterior variance, leaves the correlation at $\rho \approx 0$ throughout: weighted $\rho = 0.006$, and $\rho = 0.024$ on the most reliably measured half (Figure 3). Removing the measurement noise the reporter is entitled to does not uncover a hidden agreement.

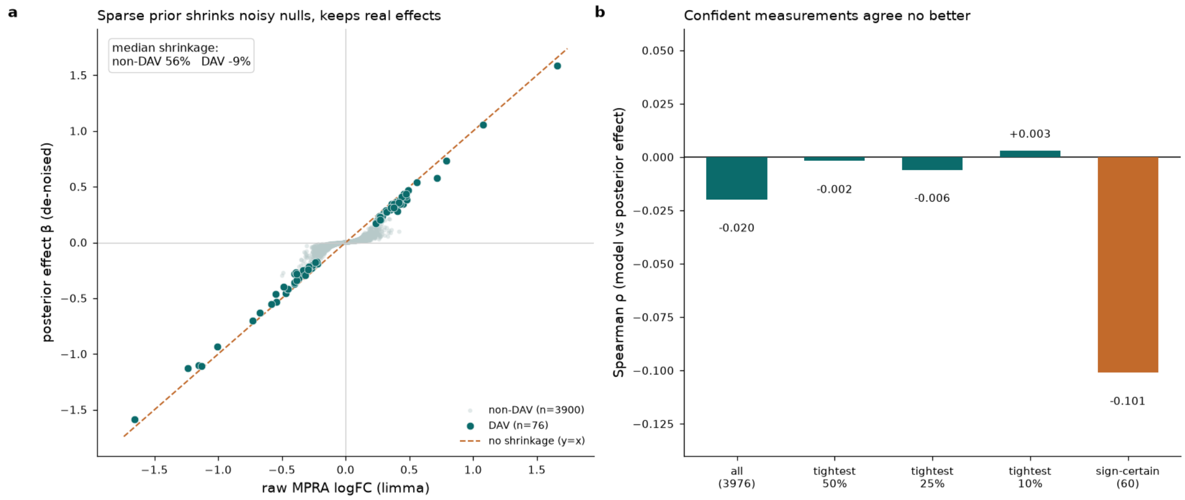

Figure 3. A from-scratch Bayesian measurement model rules out reporter noise. (a) Raw lentiMPRA effect versus horseshoe-shrunk posterior effect, colored by reliability; noisy near-null measurements are pulled toward zero (median shrinkage 56% for non-DAVs) while large real effects survive on the diagonal. (b) Model↔assay Spearman agreement on the raw set, on progressively more reliably measured subsets, and on sign-certain variants; it stays at ρ ≈ 0 everywhere.

2.3 A sequence-level cluster the reporter is absent from

Adding a second, architecturally independent foundation model turns a single-model finding into a structural one. We scored all 3,976 variants with Borzoi (524 kb windows, brain DNase/CAGE/RNA tracks) on GPU. The two models agree strongly with each other ($\rho = +0.55$, $p \approx 0$), yet Borzoi shows the same null agreement with the reporter as AlphaGenome ($\rho = -0.01$, $p = 0.52$; Figure 4a,b). The full estimator agreement matrix (Figure 4c) makes the structure explicit: AlphaGenome, Borzoi, motifbreakR and phyloP form a positive block — a sequence-level cluster — while the reporter row and column are ≈ 0 against all of them.

The model↔reporter split is therefore not an AlphaGenome-specific quirk; it is a systematic separation between what sequence models compute and what the integrated reporter measures. We are careful not to overstate the cluster: two similar sequence-to-function models agreeing is partly expected from overlapping training data, and the ties to the truly independent base-level estimators (motifbreakR ρ = 0.10–0.17; phyloP ρ = 0.01–0.04) are real but weak. We call it a sequence-level cluster, not an "endogenous-truth" cluster — and in Section 2.5 we test it against a genuinely independent endogenous readout.

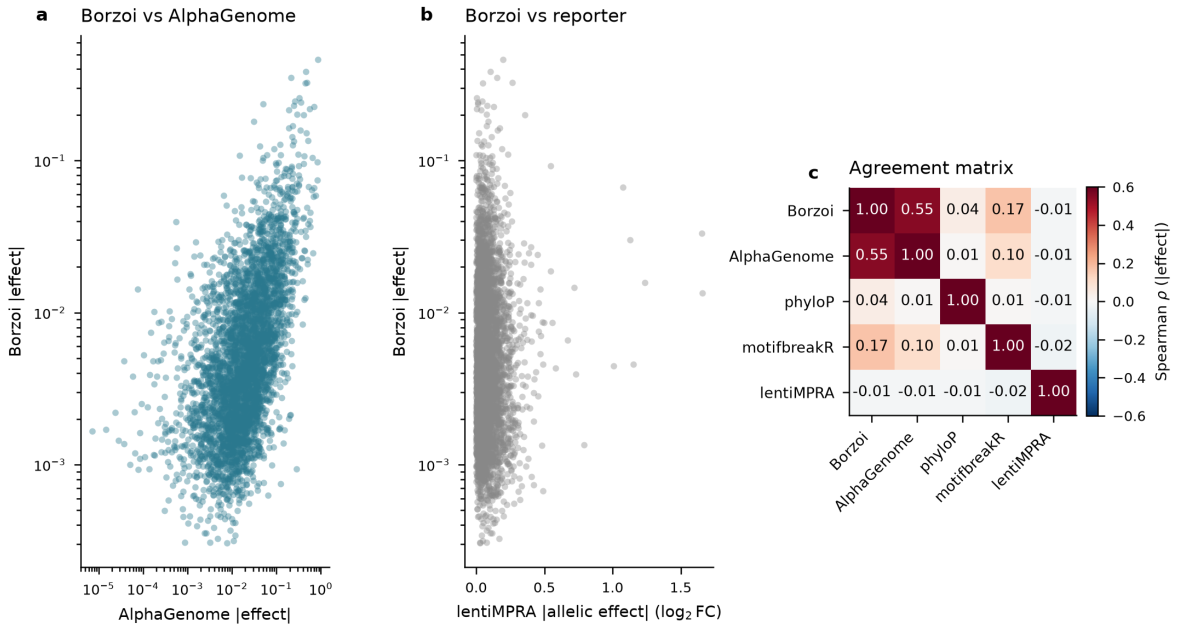

Figure 4. Two foundation models cluster together; the integrated reporter does not. (a) Borzoi versus AlphaGenome effect magnitudes agree (ρ = +0.55). (b) Borzoi versus lentiMPRA: no agreement (ρ = −0.01). (c) Full estimator agreement matrix (Spearman ρ of effect magnitude). The AlphaGenome/Borzoi/phyloP/motifbreakR block is positive; the lentiMPRA row and column are ≈ 0.

2.4 A calibrated latent-variable estimate of the agreement

Point estimates conflate shared signal with shared training-data prior, and ignore measurement noise. We therefore built a from-scratch NumPyro latent-variable generative model (Methods). Each variant carries two signed latent effects drawn jointly from a bivariate normal with unit variances and correlation ρ, where E is a latent endogenous/sequence effect and A a latent integrated-reporter effect. The endogenous sources (AlphaGenome, Borzoi, motifbreakR) load on E; the reporter loads on A; and ρ — the calibrated agreement — is identified purely through the cross-source covariance, i.e. the agreement between the two latents *after* each source's own noise is removed. phyloP is held out as orthogonal validation.

The posterior places ρ credibly at zero: ρ = −0.018, 95% HDI [−0.064, +0.027], P(ρ > 0) = 0.22 (maximum $\hat{R}$ = 1.01, no divergences; Figure 5a). The two foundation models load strongly on the shared endogenous latent (λ_{ag} = 0.87, λ_{bz} = 0.71), motifbreakR weakly (0.19), and the reporter loads on a *separate* assay latent (λ_{mp} = 0.68; Figure 5b). The per-variant latent map (Figure 5c) shows a round, untilted bulk density — the signature of ρ ≈ 0. Simulation-based calibration confirms this estimator is trustworthy (Section 2.6, Figure S3).

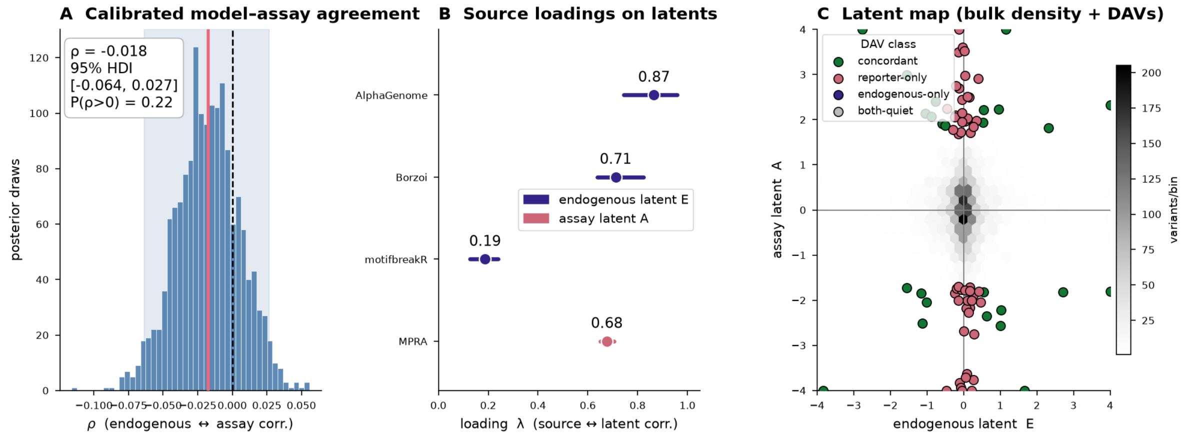

Figure 5. Calibrated latent-variable estimate of model↔assay agreement. (a) Posterior of ρ: −0.018, 95% HDI [−0.064, +0.027], credibly indistinguishable from zero. (b) Source loadings: AlphaGenome, Borzoi and motifbreakR load on the endogenous latent E; the lentiMPRA loads on a separate assay latent A. (c) Per-variant posterior map in (E, A) space; the round, untilted density is the signature of ρ ≈ 0.

2.5 An independent fine-mapped eQTL benchmark sides with the models

The analyses above establish that the models and the reporter disagree and that neither noise nor architecture explains it — but because both instruments are in dispute, neither can adjudicate the other. We therefore introduce a third, *independent* readout that privileges neither: a developmentally matched, fine-mapped fetal-cortex expression-QTL atlas [14] (SuSiE credible sets with per-variant posterior inclusion probability, PIP, and posterior effect size). Of the 3,976 variants, 370 fall in an eQTL credible set, all with a posterior effect and PIP (25 at PIP ≥ 0.5; 11 are lentiMPRA DAVs).

On this independent endogenous readout, the foundation models agree and the reporter does not. AlphaGenome's predicted effect magnitude tracks the fine-mapped eQTL posterior effect (Spearman ρ = 0.24, 95% CI [0.14, 0.33], $p = 4\times10^{-6}$), as does Borzoi (ρ = 0.28, $p = 7\times10^{-8}$), whereas the lentiMPRA does not (ρ = −0.03, 95% CI [−0.13, +0.08], n.s.; Figure 6a). The signal behaves as a genuine one should: *direction-of-effect* concordance between AlphaGenome and the eQTL rises monotonically as variants become more confidently causal — from 50% (chance) across all 370, to 60% at PIP ≥ 0.1, to 80% at PIP ≥ 0.5 (20/25, binomial p = 0.002; Figure 6b,c). The agreement is specific to the fine-mapped causal effect, not raw association strength: it is near zero against the LD-contaminated lead-SNP z-score (ρ = 0.001) and appears only against the PIP-shrunk posterior effect — concretely illustrating why lead-SNP QTLs fail as benchmarks while fine-mapped ones succeed.

We deliberately present this as *corroborating, not decisive*. It rests on 370 variants, only 25 at high PIP and 11 DAVs, so the high-confidence subset is small and the direction-of-effect result, while significant, should be read as supportive of the interpretation that the models capture native-locus regulation the decontextualized reporter does not — not as proof that the models are "right" and the reporter "wrong." It nonetheless tilts the interpretation of the model↔reporter divergence: when an orthogonal, developmentally matched, causally fine-mapped endogenous readout aligns with the models and not the reporter, the most parsimonious reading is an estimand mismatch, not model failure. This reading is reinforced by the source study itself: Deng *et al.* reported that only 55% of their differentially active variants shared direction of effect with PsychENCODE eQTLs and that the two were only weakly correlated (Pearson $r = 0.14$, $p = 0.102$; $r = 0.008$ in non-DAVs), explicitly noting the gap between reporter activity and endogenous expression. Our near-zero foundation-model↔reporter correlation is therefore not in tension with the original lentiMPRA — it recapitulates, with an independent instrument, a reporter↔endogenous discordance the source data already showed.

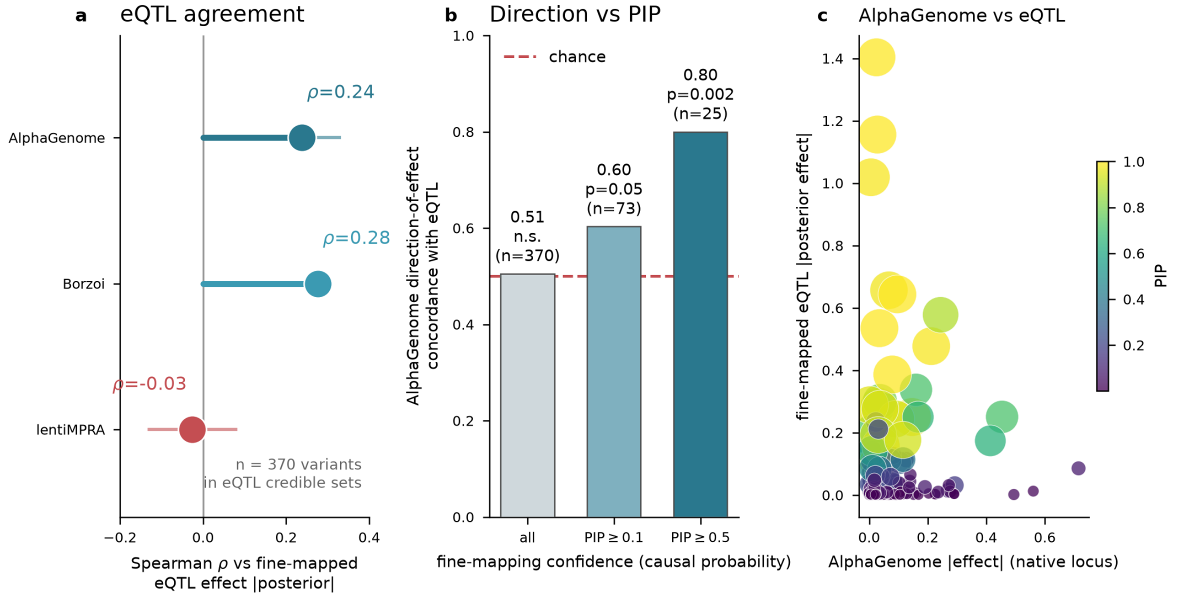

Figure 6. An independent fine-mapped fetal-cortex eQTL benchmark sides with the models. (a) Spearman agreement of effect magnitude with the fine-mapped eQTL posterior effect for AlphaGenome, Borzoi and the lentiMPRA (n = 370 variants in eQTL credible sets; bars = 95% bootstrap CI). Both models agree; the reporter does not. (b) AlphaGenome direction-of-effect concordance with the eQTL rises from chance to 80% as fine-mapping confidence (PIP) increases. (c) AlphaGenome effect vs fine-mapped eQTL posterior effect, sized/colored by PIP.

2.6 The estimator recovers known agreement (simulation-based calibration)

A reviewer's natural concern about the latent-variable ρ estimate (Section 2.4) is that its linear-Gaussian form or its priors could manufacture $\rho \approx 0$ regardless of the truth. We tested this directly by simulation-based calibration: data were generated from the fitted generative model at a grid of known ρ values, under the real heteroscedastic reporter noise (drawn from the actual limma standard errors) and the canonical loadings, then refit with the same marginalized model (Methods). The estimator recovers the true ρ with small bias and 95%-HDI coverage close to the nominal level across the grid (grid $\rho \in \{-0.3, 0, +0.3\}$; 32 fits total, with the $\rho = 0$ operating point that matches the observed finding replicated 20 times; recovered-ρ bias ≤ 0.011, 0 divergences; Figure S3). Per-ρ 95%-HDI coverage is 6/6 at $\rho = -0.3$, 17/20 (0.85) at $\rho = 0$, and 6/6 at $\rho = +0.3$ (overall 0.91); the $\rho = 0$ point is within binomial sampling error of the nominal 0.95 (95% CI for 17/20 includes 0.95), and an initial 6-replicate estimate of 4/6 rose to 0.85 once replicated. This confirms that the near-zero posterior in Section 2.4 reflects the data and not a modeling artifact.

2.7 From a verdict to a calibrated concordance map

Because the models and the reporter estimate different quantities, we deliver not a per-variant verdict but a map of where they encode the same signal. Classifying every variant by whether each latent moves yields four classes (Figure 7). Across all 3,976 variants the calibrated model assigns 626 concordant (both move), 2,003 reporter-only (reporter moves, models quiet — locally encoded / native-context-independent activity), 370 model-only (models move, reporter quiet — native-context dependent, lost in the decontextualized reporter), and 977 both-quiet. On the 76 DAVs the calibrated attribution reproduces an independent rule-based classification almost exactly (25 vs 23 concordant; 51 vs 53 reporter-only; Figure 7a), confirming the reframe is a method upgrade rather than a change of result.

Two features are worth stating. First, reporter-only dominance among DAVs is a *moderate-effect* property: of the seven DAVs with $|\log_2 FC| > 1$, four are concordant and three reporter-only, and the two single largest allelic effects are both concordant. Second, coarse annotation does not cleanly sort concordant from reporter-only DAVs (Figure 7b) — a hard annotation threshold cannot separate what a calibrated soft model can, which is why the per-variant map is presented as triage, not adjudication.

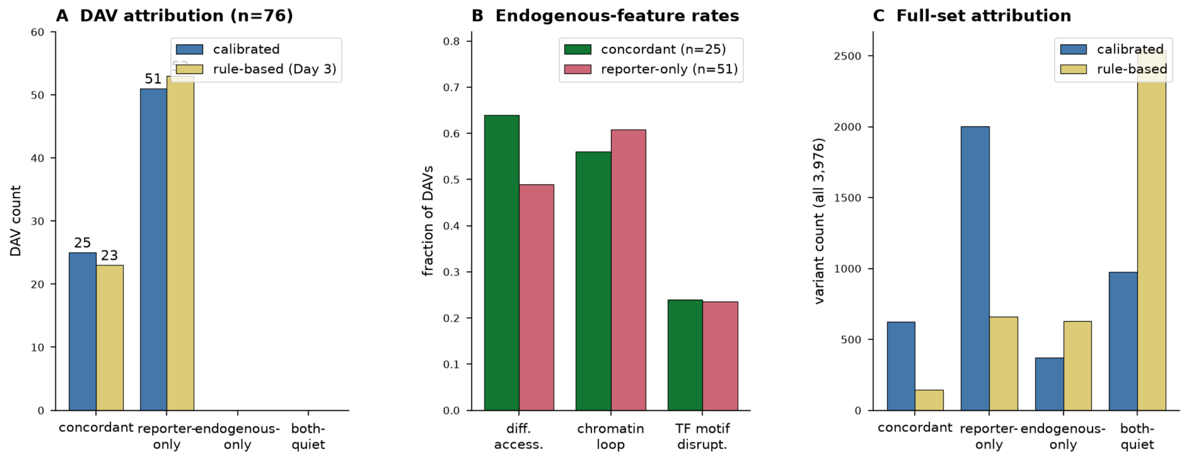

Figure 7. Calibrated concordance map. (a) Calibrated versus rule-based attribution of the 76 DAVs (25 vs 23 concordant; 51 vs 53 reporter-only). (b) Endogenous-feature rates by DAV class: accessibility, chromatin loop and TF-motif disruption do not cleanly separate concordant from reporter-only DAVs. (c) Full-set attribution across all 3,976 variants.

2.8 Near-zero concordance is uniform, and the endogenous agreement is specific

Two stratified analyses guard against the concern that a near-zero global correlation masks meaningful regimes, or that the endogenous agreement is a generic sequence-property artifact. First, model↔reporter concordance stays at $\rho \approx 0$ within every stratum of conservation, baseline enhancer activity, motif-disruption burden and QTL-vs-GWAS ascertainment (Figure 8a,c): no subgroup rescues the agreement. Second, the model↔eQTL agreement (Section 2.5) survives partial correlation controlling for phyloP conservation, baseline activity and the reporter effect (AlphaGenome $\rho = 0.24 \rightarrow 0.23$; Borzoi $0.28 \rightarrow 0.27$; Figure 8b) — so the endogenous signal is specifically the eQTL effect the models track, not GC content or sequence complexity. In domain, AlphaGenome's accessibility head discriminates caQTL overlap (AUROC 0.76, n = 36) and its contact-map head discriminates chromatin-loop membership (AUROC 0.64, n = 2,502), and base-level conservation separates the model-supported divergent class (Mann–Whitney $p = 0.002$; Figures S1, S2) — consistent with the models recovering real endogenous chromatin effects, without resting the claim on any single line of evidence.

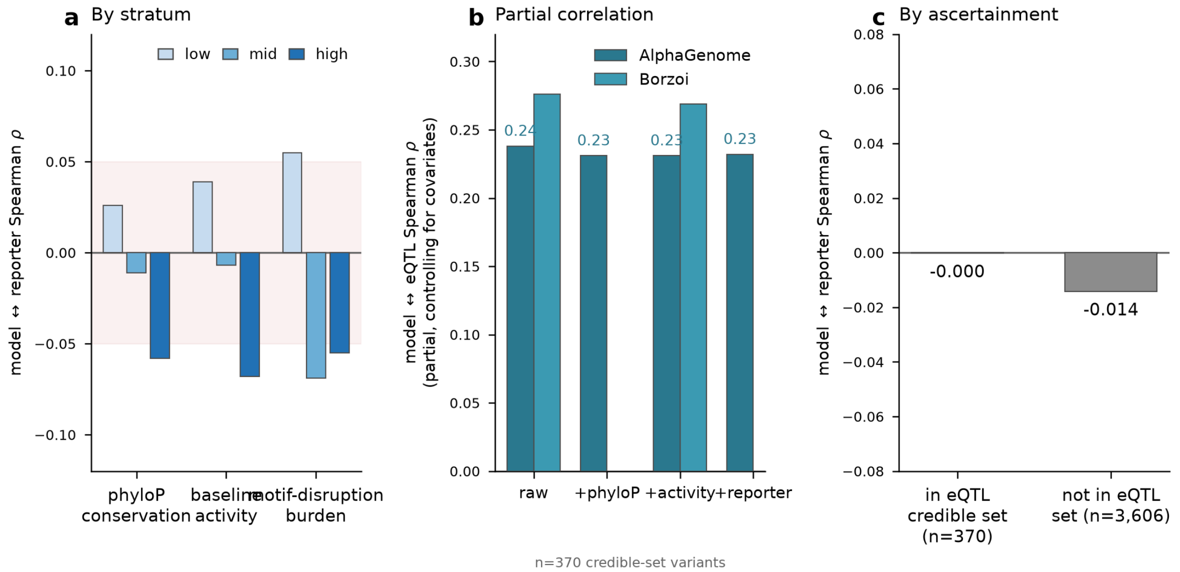

Figure 8. Near-zero model↔reporter concordance is uniform; endogenous agreement is specific. (a) Model↔reporter Spearman ρ within tertiles of conservation, baseline activity and motif burden — all ≈ 0. (b) Model↔eQTL agreement survives partial correlation controlling for conservation, activity and reporter effect. (c) Model↔reporter ρ is unchanged whether or not variants are in an eQTL credible set (ascertainment-independent).

2.9 Reporter potential does not propagate predictably to the endogenous effect

The eQTL benchmark shows the reporter misses the endogenous signal, but not *why*. One mechanistic hypothesis is attractive: the lentiMPRA measures intrinsic regulatory *potential*, and that potential reaches endogenous expression only where the native context — accessibility, enhancer–promoter contact, transcription-factor availability — permits it. If so, reporter and endogenous effects should agree specifically at context-favorable loci, and a reporter×context interaction should predict the endogenous effect the reporter alone cannot. We tested this directly with an Activity–Context–Propagation (ACP) model, built from scratch on the 370 fine-mapped variant–gene pairs, with the SuSiE posterior effect [20] as the endogenous outcome (its posterior SD as known measurement error, PIP as causal-confidence weight), the oriented reporter effect as predictor, and provenance-controlled context axes (measured cortical ATAC, activity-by-contact enhancer–promoter contact, phyloP, motif disruption, enhancer redundancy) reduced to orthogonal composites; AlphaGenome/Borzoi tracks were held out of the primary context axes so a benchmarked model is not used as its own context.

The hypothesis is not supported at this sample size, and the negative is informative (Figure 9). In repeated PIP-weighted nested cross-validation, no model in the ladder — reporter-only, context-only, additive, or interaction — beat an intercept null on held-out effect, and the crucial comparison, the reporter×context interaction against the additive model, was a coin flip (interaction better in 47% of paired folds; mean $\Delta R^2 = -0.03$). A from-scratch Bayesian hierarchical version (NumPyro; regularized-horseshoe shrinkage, PIP-tempered likelihood, and an explicit intrinsic-scatter term beyond the known eQTL measurement error) agrees: the interaction coefficient is centered at zero ($\gamma = 0.001$, 95% CI [−0.056, +0.068], $P(\gamma>0) = 0.51$), the reporter main effect includes zero ($\beta_M = 0.06$, 95% CI [−0.15, +0.26]), and the fitted intrinsic scatter ($\sigma_{extra} = 0.16$, comparable to the median measurement noise) dominates the explainable variance — endogenous allelic effect is largely variance that neither reporter potential nor measured context accounts for. Omitting the intrinsic-scatter term inflates an apparent reporter effect by forcing all variance through the small eQTL measurement error; the better-specified model, with calibrated residuals (SD 1.6 vs 6.7), removes it.

Classifying every variant by whether each instrument moves — with a posterior-based endogenous-null definition, P(|Y| < δ | data) > 0.8 rather than a non-significance threshold — reproduces the reporter-only dominance in mechanistic terms: the "latent-potential" class (reporter-active, endogenously silent; n = 34) is by far the largest active class, against 4 propagating (both move) and 14 context-only. Most reporter-active variants are endogenously silent. But this class is not explained by the measured context: against reporter-matched controls (matched on effect magnitude, reporter uncertainty, PIP, activity and conservation), no context axis significantly predicts latent-potential membership (accessibility, coupling, redundancy and conservation odds ratios all ≈ 0.9, 95% CIs crossing 1, all p > 0.7). The odds-ratio directions match the hypothesis (latent variants trend toward lower accessibility and weaker contact) but are far from significant at n = 34. The ACP analysis thus localizes the reporter↔endogenous divergence to a concrete, dominant class of variants without, at this sample size, attributing it to the specific context axes measured.

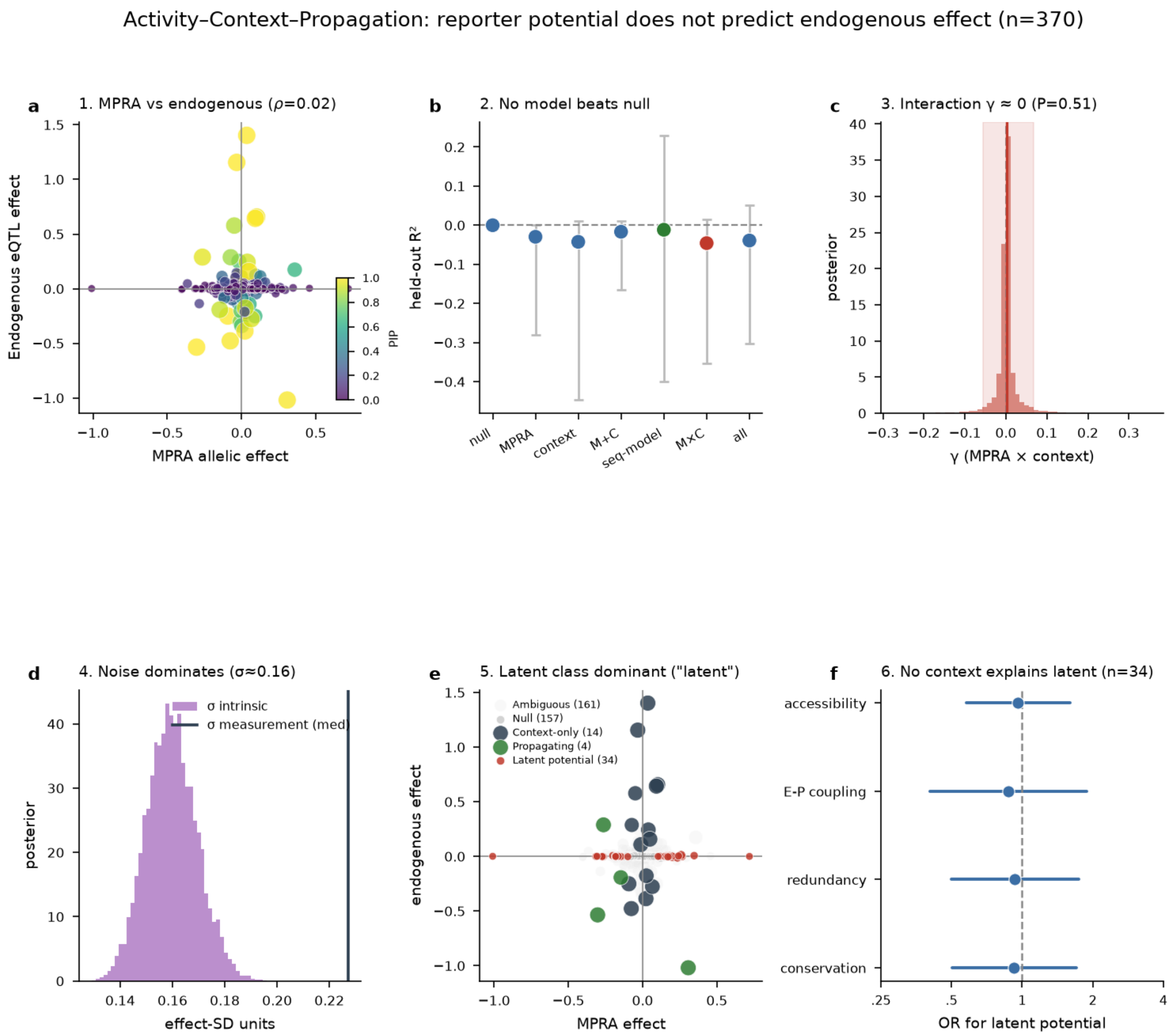

Figure 9. Activity–Context–Propagation model: reporter potential does not propagate predictably to the endogenous effect (n = 370). (1) Reporter effect does not predict the fine-mapped eQTL effect (ρ = 0.02). (2) Nested PIP-weighted cross-validation: no model beats the null; the reporter×context interaction (red) does not help. (3) Bayesian interaction coefficient γ is centered at zero. (4) Fitted intrinsic scatter is comparable to the median measurement noise and dominates the explainable variance. (5) 2×2 propagation classes: the latent-potential class (reporter-active, endogenously silent) is the largest active class. (6) No context axis significantly predicts latent-potential membership against matched controls (all odds-ratio CIs cross 1).

2.10 Context type, not context-window length, separates the reporter from the models

The eQTL benchmark (Section 2.5) and the ACP analysis (Section 2.9) both show the sequence models track the endogenous effect while the reporter does not. A natural question is whether that gap is about the *amount* of genomic context a model integrates — whether a model reading only a short, reporter-sized window would behave like the reporter. It does not: the divide is context *type* (native locus vs decontextualized element), not context *length*. Placing the estimators on a context ladder against the fine-mapped eQTL anchor (Figure 10a), AlphaGenome tracks the endogenous effect essentially identically at a 100 kb and a 1 Mb window (Spearman $\rho = 0.23$ vs 0.22; the two scorings are near-interchangeable, $\rho = 0.93$ on this set and $\rho = 0.99$ genome-wide, Methods), Borzoi tracks it at $\rho = 0.28$, and the lentiMPRA does not track it at any setting ($\rho = -0.03$, n.s.). The estimator-proximity matrix (Figure 10b) makes the point structurally: the two AlphaGenome windows and Borzoi form a positive block ($\rho = 0.55$–0.93), while the reporter is near-orthogonal to all of them ($\rho \approx 0$). Enlarging or shrinking a model's receptive field within its native-context regime barely moves its agreement with endogenous genetics; removing native context altogether — which is what the integrated reporter does — is the step that abolishes it. This localizes the estimand mismatch to the presence or absence of native genomic context rather than to context extent.

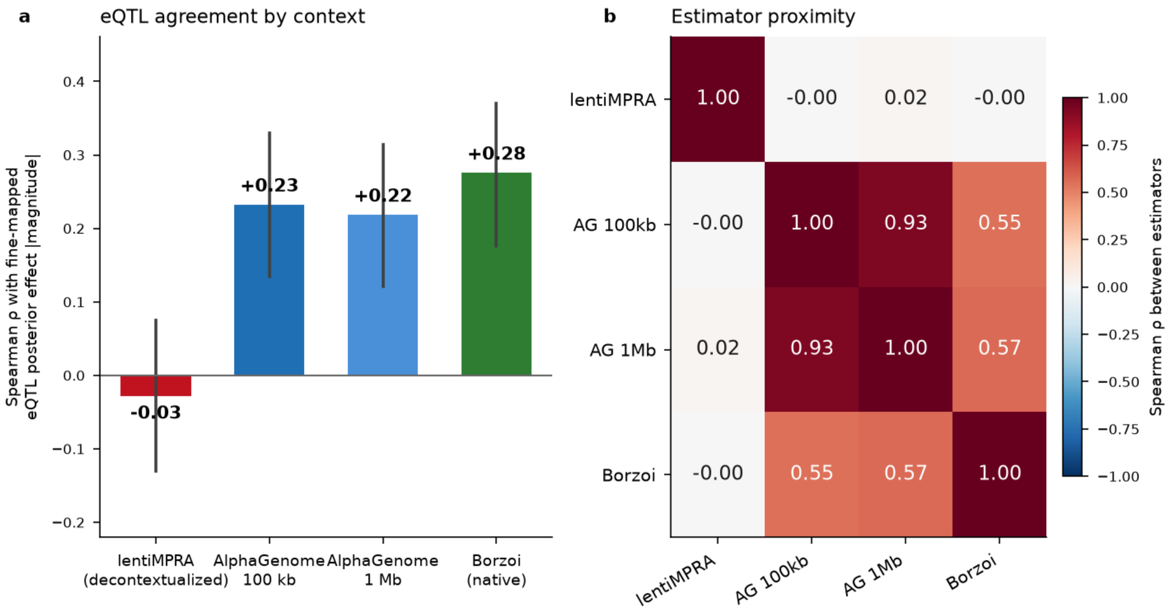

Figure 10. Context type, not context-window length, separates the reporter from the models. (a) Spearman agreement of effect magnitude with the fine-mapped eQTL posterior effect for the lentiMPRA, AlphaGenome at a 100 kb and a 1 Mb window, and Borzoi (n = 369 credible-set variants; error bars = 95% bootstrap CI). The two AlphaGenome windows are near-identical and both, with Borzoi, track the eQTL; the reporter does not at any setting. (b) Estimator-proximity matrix (Spearman ρ of effect magnitude): the two native-context windows and Borzoi cluster ($\rho = 0.55$–0.93) while the reporter row/column are ≈ 0.

2.11 Case study of the concordant core: direction resolution and variant prioritization

The concordance map defines *where* the two instruments encode the same signal, but the concordance label is computed on effect *magnitude* — it records that both latents move, not that they move the *same way*. For functional follow-up, the direction of the allelic effect is what matters, so we examined the sharpest slice of the map: the 25 concordant, differentially active variants (the concordant core of the 76 DAVs), which include the two largest allelic effects in the library. For each we resolved the signed direction of effect across all available estimators —

lentiMPRA, AlphaGenome, Borzoi, and, where the variant intersects a fine-mapped credible set, the developing-cortex eQTL — and ranked the 25 by a transparent, equally-weighted convergent-evidence score over seven axes (direction agreement, reporter effect and significance, fine-mapped eQTL support, psychiatric-GWAS linkage-disequilibrium, TF-motif disruption, brain regulatory activity, and conservation; Methods).

Magnitude concordance is only weakly predictive of direction concordance (Figure 11a). Of the 25, only 10 show full three-way direction agreement between the reporter and both foundation models; in 11 the two models jointly oppose the reporter's sign, and 4 are two-of-three splits. This is consistent with the calibrated $\rho \approx 0$ at the whole-set level: even within the concordant core, where both instruments register an effect, they agree on *which* allele is up only about as often as chance would give among the strongest reporter effects. The ranking (Figure 11b,c) therefore promotes the variants that combine direction agreement with orthogonal support. The top candidate, rs76980967 (chr15:88,955,123 A>G), is the only variant in the set where all four estimators agree in direction: reporter, AlphaGenome, and Borzoi all place the alternate allele as activity-increasing, and it sits in a fetal-cortex eQTL credible set (PIP = 0.64) whose posterior effect points the same way, nominating *MFGE8* as the target gene. The next ranked candidates carry cell-type-resolved enhancer–gene support (rs2154984, an ABC enhancer contacting *STIP1* across seven cortical cell types, in LD with a bipolar-disorder signal) or TF-motif disruption at psychiatric-GWAS loci (rs62485366 disrupting a VEZF1 motif; rs139376573 an OSR2 motif).

To interpret the concordant core against the classes it is drawn from, we assembled a matched 20-variant panel: the 8 top-ranked concordant variants against 4 model-only, 4 reporter-only, and 4 both-quiet variants (matched on baseline reporter activity; Methods). The panel reproduces the whole-set structure in miniature (Figure 12). Model-only variants carry large sequence-model effects with a quiet reporter (mean |AlphaGenome| = 0.64 vs. |MPRA| = 0.02), reporter-only variants the reverse (|MPRA| = 1.13 vs. |AlphaGenome| = 0.02), and both-quiet controls sit at the origin. The concordant group is the only one in which both instruments move together — and even there, the direction resolution above shows the agreement is on magnitude more than on sign. The clearest functional experiment this prioritization motivates is CRISPR base editing at the native locus in neural progenitors or cortical organoids, reading out the nominated target gene: the strongest single candidate is rs76980967→*MFGE8*, the one variant that already satisfies reporter, model, and fine-mapped-eQTL direction agreement and whose edit has a pre-registered directional prediction (alternate allele increases *MFGE8*).

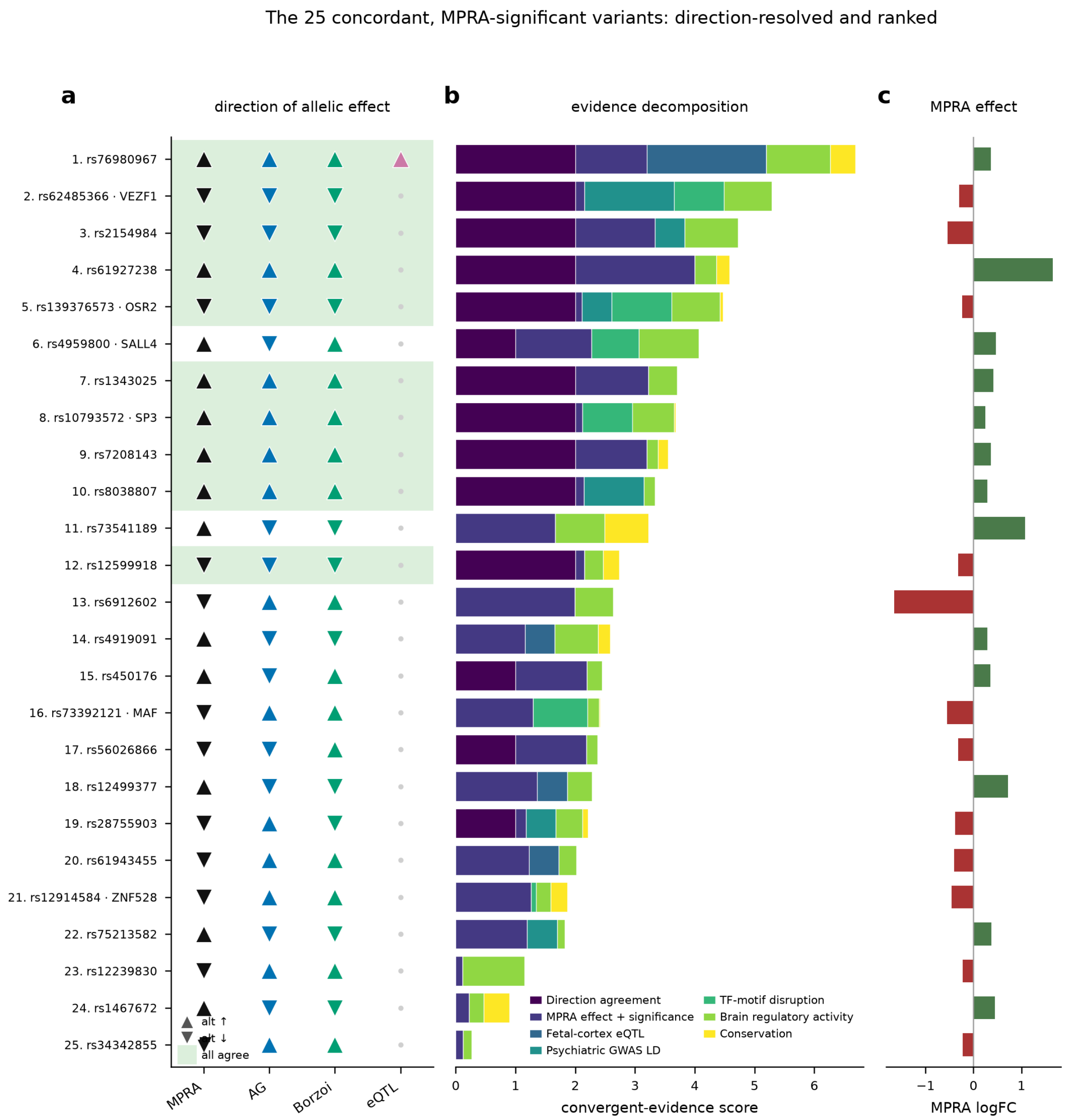

Figure 11. Case study of the 25 concordant, differentially active variants: direction resolution and ranking. (a) Signed direction of the allelic effect per estimator (up-triangle, alternate allele increases activity; down-triangle, decreases). Rows are ranked by convergent-evidence score; the green band marks the 10 variants where lentiMPRA, AlphaGenome and Borzoi all agree in direction. The eQTL column is filled only for the one variant in a fine-mapped credible set with PIP ≥ 0.1 (rs76980967). Only 10 of 25 magnitude-concordant variants are direction-concordant. (b) The convergent-evidence score decomposed into its seven equally-weighted axes. (c) lentiMPRA signed effect ($\log_2$ fold change), confirming all 25 are genuine reporter moves — the disagreement in (a) is about sign, not detection.

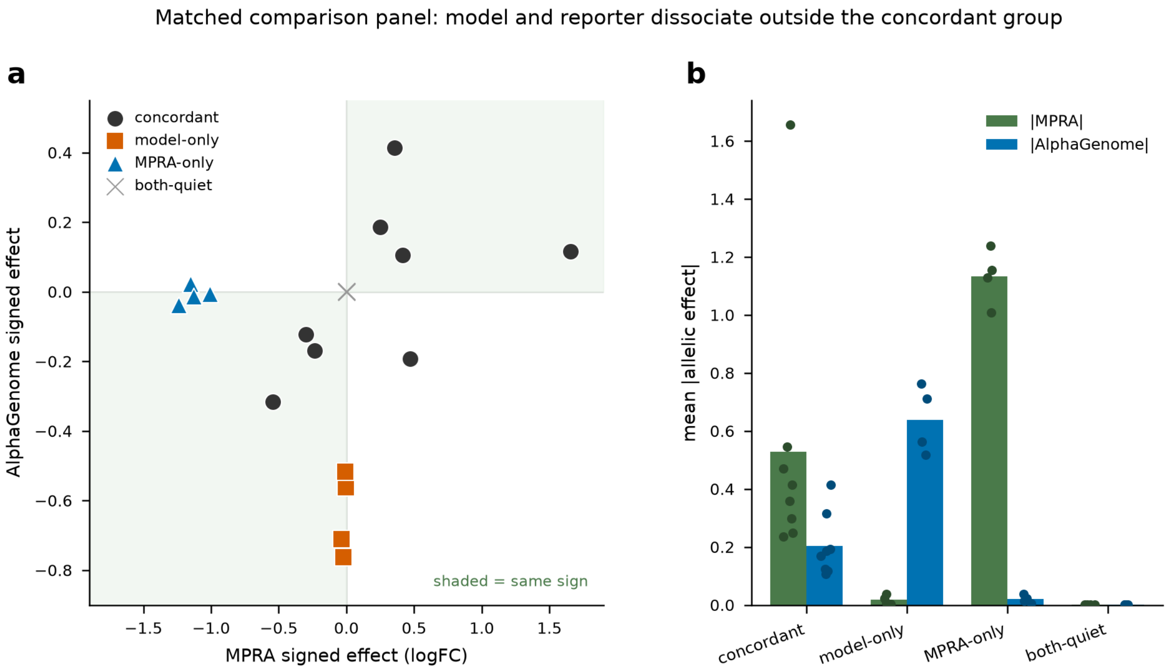

Figure 12. Matched comparison panel: model and reporter dissociate outside the concordant group. A 20-variant panel of 8 top-ranked concordant variants against 4 model-only, 4 reporter-only, and 4 both-quiet variants. (a) AlphaGenome versus lentiMPRA signed effect; shaded quadrants mark sign agreement. The concordant variants lie predominantly in the same-sign quadrants, while model-only (large model effect, quiet reporter) and reporter-only (large reporter effect, quiet model) variants occupy the axes. (b) Mean absolute allelic effect by group for the reporter and AlphaGenome, with per-variant points overlaid: the two instruments dissociate cleanly outside the concordant group.

2.12 Translational outlook: from a direction-resolved locus to a druggable target

A direction-resolved variant is only useful downstream if it points at something actionable. Because these are regulatory variants, the drug target is not the variant but the gene it controls, and the therapeutic hypothesis is set by the direction of effect — which the case study resolves. We therefore traced each concordant variant through a five-step chain (Figure 13a): regulatory variant → causal gene and direction (delivered by the triangulation) → target tractability → disease sense and safety → functional validation. Of the 25, nine carry a confidently linked gene (eQTL, activity-by-contact enhancer–gene contact, or a disrupted transcription-factor motif), yielding 13 candidate target genes; the remaining 16 have no confidently linked gene at this resolution and are not yet actionable.

We scored the 13 genes for druggability using Open Targets tractability, genetic constraint, and disease association (Figure 13b; Methods). None has an approved drug or clinical candidate — every one is a novel target, as expected for common regulatory psychiatric risk. Ranking on a composite of causal-pointer strength, direction agreement, druggable modality, and psychiatric-disease relevance places two genes clearly ahead. *STIP1* (rs2154984) is the strongest overall candidate: a strong enhancer–gene link across seven cortical cell types, full direction agreement, a genome-wide bipolar-disorder association, and small-molecule tractability (a ligand-bound structure) — tempered by a low loss-of-function tolerance (LOEUF 0.31), implying dosage sensitivity. *MFGE8* (rs76980967) is the strongest by causal confidence — the only variant with four-estimator direction agreement and a fine-mapped eQTL — and encodes a secreted, antibody-tractable, loss-of-function-tolerant (LOEUF 1.04) protein, but its disease anchor is weaker (a gene-level schizophrenia association rather than a genome-wide signal at the locus). The transcription-factor targets (*VEZF1*, *SP3*, *MAF*) carry the best disease evidence but only degrader-pathway tractability, and *RPA1* has the best small-molecule pocket but no psychiatric relevance.

Asking what disease these variants implicate exposes a second, honest limit (Figure 14). Only seven of the 25 carry a nameable psychiatric indication: six in linkage disequilibrium with a genome-wide psychiatric signal (bipolar disorder, schizophrenia, the PGC cross-disorder analysis, major depression, autism) and one sub-threshold (neuroticism). Of the remainder, eight are linked only through brain molecular-QTL credible sets with no disease GWAS at the locus — including *MFGE8* — six sit under non-psychiatric GWAS traits, and four have no credible set in Open Targets. This is the expected profile of variants discovered by function rather than by association: mechanism-strong but indication-provisional. It cleanly separates the disease-anchored candidates (*STIP1*, bipolar disorder; *VEZF1*, bipolar disorder and schizophrenia) from the mechanism-anchored one (*MFGE8*), and it argues that variant-to-indication assignment, not just target tractability, is the rate-limiting step for turning a concordance map into a therapeutic hypothesis. In all cases an endogenous, allele-specific perturbation at the native locus remains the gate: tractability and disease association are worth acting on only once the causal gene and direction are confirmed. The most direct such experiment tests the estimand-mismatch hypothesis itself — CRISPR base editing (or prime editing) of representatives from all four calibrated classes (model-and-reporter concordant, model-only, reporter-only, and both-quiet), followed by endogenous target-gene expression and chromatin readout in neural progenitors or cortical organoids. If model-only variants move endogenous expression while reporter-only variants do not, that is direct causal confirmation that the two instruments estimate different quantities; the concordant core simply names the variants where a single edit tests the cleanest prediction first.

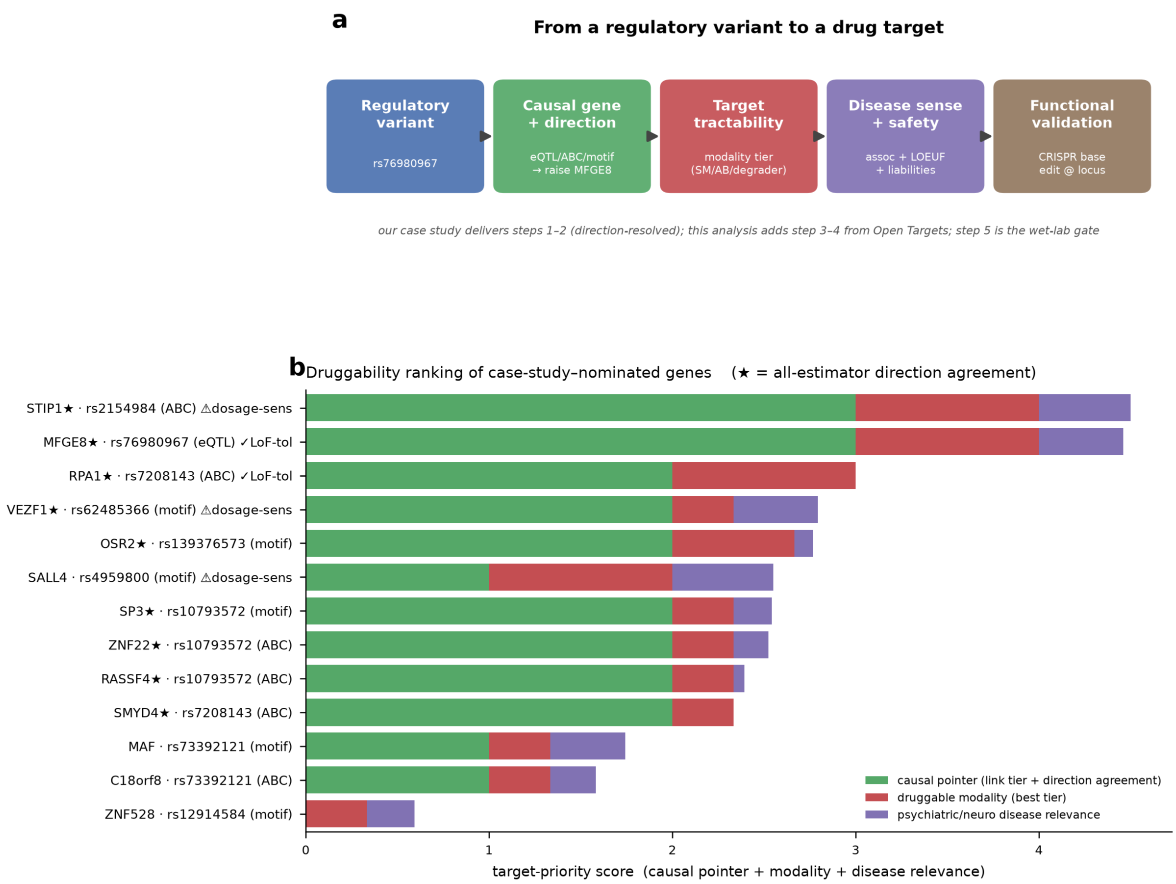

Figure 13. From a regulatory variant to a druggable target. (a) The five-step chain from a direction-resolved regulatory variant to a functionally validated drug target; the case study delivers steps 1–2, this analysis adds steps 3–4 from Open Targets, and step 5 is the wet-lab gate. (b) Druggability ranking of the 13 genes nominated by the concordant core, decomposed into causal-pointer strength (link tier + direction agreement), best druggable modality tier, and psychiatric/neurological disease relevance. Stars mark all-estimator direction agreement; annotations flag loss-of-function-tolerant (safer to modulate) versus dosage-sensitive genes. No gene has an approved or clinical-candidate drug.

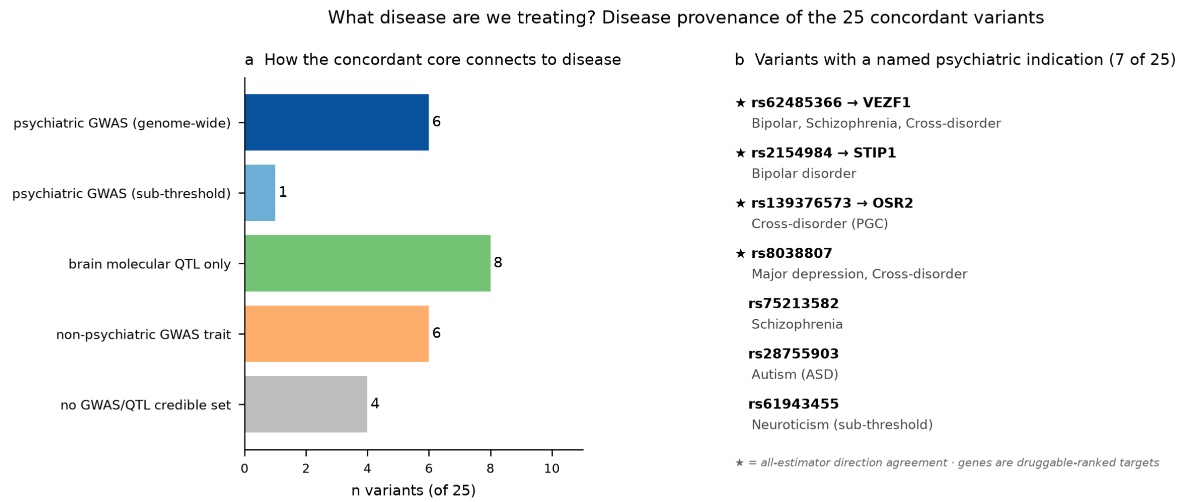

Figure 14. Disease provenance of the concordant core. (a) How the 25 variants connect to disease: six are in linkage disequilibrium with a genome-wide psychiatric signal, one is a sub-threshold psychiatric hit, eight are linked only through brain molecular-QTL credible sets, six sit under a non-psychiatric GWAS trait, and four have no Open Targets credible set. (b) The seven variants with a nameable psychiatric indication, with their nominated druggable target where available. Stars mark all-estimator direction agreement.

2.13 Robustness of the core results to reviewer-raised alternatives

We stress-tested the three load-bearing results against the modelling choices most likely to drive them. First, the measurement-error denoiser treats the limma-moderated standard errors as known; re-fitting the hierarchical shrinkage model while scaling every reporter standard error by ×0.5 and ×2 leaves the denoised model↔reporter agreement essentially unchanged (Spearman ρ = −0.019, −0.019 and −0.019 at ×0.5, ×1, ×2, versus −0.020 undenoised; Figure 15a), so the near-zero conclusion does not rest on taking the SEs as exact. Second, the latent-variable model assigns each source a single-factor loading; freeing the reporter to load additionally on the endogenous latent (a cross-loading λ_x) returns a posterior indistinguishable from zero (λ_x = −0.02, 95% credible interval [−0.26, +0.23]) while both sequence models retain strong endogenous loadings ($\lambda \approx$ 0.78–0.80; Figure 15b), so the calibrated near-zero agreement is not an artefact of the single-loading restriction. Third, the direction-of-effect concordance at high causal confidence (20 of 25 at PIP ≥ 0.5) has a bootstrap 95% confidence interval of [0.64, 0.96] and survives a sign-flip permutation test that preserves allele orientation (p = 0.003, concordant with the binomial p = 0.004; Figure 15c). All three analyses reinforce the original conclusions rather than qualifying them.

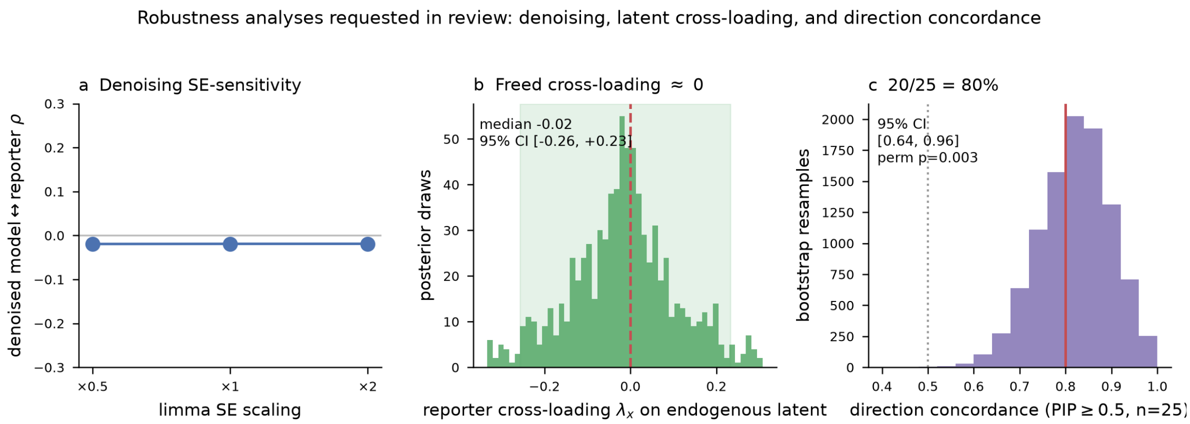

Figure 15. The core results are robust to reviewer-raised modelling alternatives. (a) Denoised model↔reporter Spearman ρ as the limma standard errors are scaled ×0.5 to ×2; the value stays at ≈ −0.02. (b) Posterior of the freed reporter cross-loading λ_x on the endogenous latent, centred on zero. (c) Bootstrap distribution of the PIP ≥ 0.5 direction-of-effect concordance (n = 25); red line, observed 0.80; dotted line, chance; permutation p = 0.003.

2.14 A third independent model (Enformer) reproduces both halves of the result

The panel so far rests on two models (AlphaGenome, Borzoi) that, while independently trained, share functional-genomics training data of similar kind. To test whether the estimand mismatch generalizes beyond this pair, we scored all 3,976 variants with Enformer — a third sequence model with public weights and a distinct training regime (18 min on one GPU; all variants scored). Enformer reproduces *both* halves of the result. Its effect magnitudes agree with AlphaGenome (ρ = +0.58) and Borzoi (ρ = +0.81, as expected given the shared Basenji lineage), yet show the *same* null agreement with the integrated reporter as the other two models (ρ = −0.01, p = 0.47; Figure 16a). And against the independent fine-mapped eQTL benchmark, Enformer tracks the endogenous effect at essentially the same strength as AlphaGenome and Borzoi (ρ = 0.235, p = $5{\times}10^{-6}$, n = 370; Figure 16b) — three models, one endogenous estimand. The mismatch is therefore a systematic property of sequence-to-function models relative to this reporter, not a quirk of a single architecture or training set.

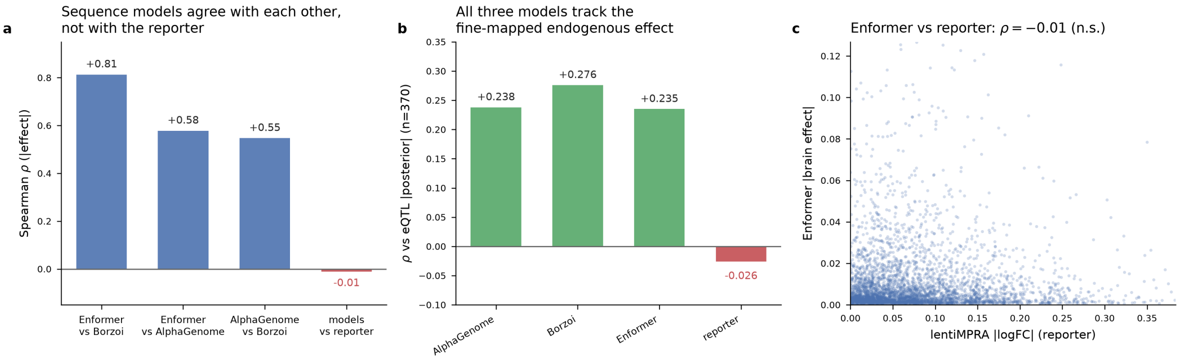

Figure 16. A third sequence model (Enformer) reproduces the estimand mismatch. (a) Pairwise agreement of effect magnitude: all three sequence models agree with one another (Spearman ρ = 0.55–0.81, blue) but none agrees with the integrated reporter (ρ ≈ 0, red). (b) Agreement with the fine-mapped eQTL posterior effect (n = 370): all three models track the endogenous effect at ρ ≈ 0.24–0.28 (green) while the reporter does not (red). (c) Enformer brain-track effect vs lentiMPRA |logFC| across all 3,976 variants: no relationship (ρ = −0.01, n.s.).

3. Discussion

We asked how reliably DNA foundation models predict single-base regulatory variant effects and, when they disagree with an MPRA, whether the disagreement is model error, measurement noise or real biology. The answers are now consistent. Model and reporter effect sizes do not agree; the disagreement is not reporter measurement noise (Bayesian denoising leaves $\rho \approx 0$); the two foundation models agree with each other and weakly with independent base-level estimators; and a generative latent-variable model, validated by simulation-based calibration, puts the calibrated model↔reporter agreement credibly at zero ($\rho = -0.018$, HDI [−0.064, +0.027]).

The most informative new result is the external anchor. Against an independent, developmentally matched, fine-mapped fetal-cortex eQTL atlas, the models' predictions track the endogenous effect ($\rho = 0.24$–0.28) while the reporter does not, and the models' direction-of-effect concordance climbs to 80% on confidently causal variants. Because this readout privileges neither instrument in dispute, it converts a symmetric "two instruments disagree" into an asymmetric observation: the instrument that also agrees with independent endogenous genetics is the sequence model, not the decontextualized reporter. We are deliberate about the strength of this claim — it rests on 370 variants (25 at PIP ≥ 0.5) and is corroborating rather than decisive — but it is the piece of evidence that most directly discriminates the estimand-mismatch reading from a model-failure reading.

Under the estimand-mismatch reading, the divergence is an expected consequence of two different measurement targets, not a failure of either instrument. A foundation model predicts activity in the native genomic locus, complete with native distal enhancers, 3D contacts and position effects; the lentiMPRA measures a short oligo chromatinized at ectopic integration sites, retaining chromatin but not native context. The reporter-only class then captures locally encoded, position-independent activity, and the model-only class captures native-context-dependent activity the decontextualized reporter cannot reconstruct. Near-zero global agreement is compatible with, and supported by, an estimand mismatch when most variants' effects are context-sensitive; we present it as consistent with that mechanism rather than as proof that mismatch is the sole cause. Consistent with this, the divergence is specifically about context *type* rather than context *extent*: a model's agreement with endogenous genetics is essentially unchanged between a 100 kb and a 1 Mb receptive field but is abolished by the reporter's removal of native context altogether (Section 2.10), so the operative variable is the presence of the native locus, not the length of sequence a predictor reads.

This work is complementary to, and distinct from, the recent literature stress-testing sequence models. Two Nature Genetics 2023 studies are the closest neighbors: Huang et al. showed cross-individual transcriptome variation is poorly explained by current models (performance near zero across 421 Geuvadis individuals) [17], and Sasse et al. found models frequently mispredict the direction of personal expression-variant effects across 839 ROSMAP individuals [18]. Both document a performance gap against endogenous expression — the correct estimand, imperfectly powered. Our contribution is orthogonal in kind: we show a benchmark can be mis-specified at the level of the *estimand* when the reference is a decontextualized reporter, and we provide a calibrated latent-variable framework plus an independent fine-mapped QTL arbiter. The broader model literature — Enformer [2], Sei [15], Orca [16] and cell-type-aware extensions such as Murphy et al. [19] — has advanced mainly by scale, resolution and mechanistic breadth; our work is deliberately evaluative and complementary, suggesting that the choice of benchmark target matters as much as the model.

The practical implication is a recommendation: to benchmark *in silico* regulatory variant-effect prediction, a fine-mapped, developmentally matched endogenous QTL is a more appropriate arbiter than a decontextualized reporter, and cross-instrument comparisons should be read as concordance maps between imperfect estimators rather than as model-versus-truth scoreboards.

What the fine-mapped benchmark does *not* deliver is a mechanism for the divergence. We tested the most natural one — that reporter-measured potential propagates to endogenous expression specifically where native context is favorable — with the Activity–Context–Propagation model, and it is not supported at this sample size: neither the reporter nor a reporter×context interaction predicts held-out endogenous effect, and an explicit intrinsic-scatter term absorbs most of the variance. The mechanistic content that survives is descriptive but sharp: reporter-active, endogenously silent "latent-potential" variants are the dominant active class, so the divergence is concentrated rather than diffuse — we simply cannot yet attribute it to accessibility, contact or conservation with 34 such variants. Powering that attribution — not the availability of a causal-base benchmark — is now the rate-limiting step, and the direct remedy is a hierarchical model over several matched MPRA–eQTL/CRISPR datasets across cell types, with dataset and tissue as random effects.

Limitations

The fine-mapped eQTL tie-breaker and the ACP analysis rest on 370 credible-set variants (73 at PIP ≥ 0.1, 25 at PIP ≥ 0.5), and the reporter-active/endogenously-silent class that carries the mechanistic signal numbers only 34; the ACP null is therefore an absence of *detectable* context-dependence, not evidence of its absence, and the context enrichment is underpowered. At this sample size the multiplicative ACP product form is not identifiable against additive-on-log alternatives, so we treat it as a regularization target rather than a hypothesis we adjudicate; the posterior-based endogenous-null threshold ($\delta = 0.05$ on the SuSiE posterior scale) is a modeling choice. The single-dataset n, not the availability of a causal-base readout, is the binding constraint on mechanism.

(i) The benchmark comprises 3,976 biallelic variants with 76 DAVs after QC. The source study designed 17,069 variants, of which 15,335 passed both-allele QC and 8,029 showed activity from at least one allele, calling 164 DAVs across the full (including indel and multi-allelic) library; restricting to biallelic SNVs scoreable by both foundation models yields our 3,976 / 76. This filtering removes variant classes the sequence models cannot score rather than selectively enriching for agreement, but the reduction is substantial and we report the full funnel (17,069 → 15,335 → 8,029 → 3,976) so the exclusions are transparent. (ii) The eQTL anchor, though independent and fine-mapped, is modest in size (370 overlapping variants, 25 at PIP ≥ 0.5, 11 DAVs); the direction-of-effect result is significant but should be read as supportive. (iii) A second, controlled-access neuronal MPRA (Zeng et al.) could not be integrated because its variant-level results are not openly available; we document this rather than gate the analysis behind an access application. (iv) Model brain tracks are largely adult/bulk while the assay and eQTL anchor are fetal; developmental-stage mismatch remains a live secondary factor. (v) That two similar sequence models agree is partly expected from overlapping training data; the latent-variable model isolates cross-source signal but cannot fully remove a shared prior. (vi) The latent generative model is linear-Gaussian; it captures monotone shared signal, not nonlinear or sign-flipping relationships. (vii) The AlphaGenome and Borzoi summaries pool brain/CNS tracks and are robust to aggregation ($\rho \geq 0.98$), but remain summaries of high-dimensional predictions. (viii) Three extensions we regard as important could not be run in the present environment and are stated as future work rather than claimed. We benchmark three sequence models (AlphaGenome, Borzoi, Enformer; Section 2.14); adding further architecturally distinct predictors of a different paradigm (DeepSEA, deltaSVM) would further test whether the sequence-level cluster and its divergence from the reporter persist across training regimes, but requires additional GPU inference and model weights we did not have access to. We use a single lentiMPRA; replication on an independent assay spanning episomal and integrated designs and alternative promoter contexts would directly localize the estimand mismatch to delivery versus promoter context, and is the single most valuable external validation. And because AlphaGenome and Borzoi are trained on functional-genomics compendia that may include brain tissues, part of the model↔eQTL agreement could reflect shared tissue signal rather than fully independent validation; we cannot audit their training corpora from published descriptions alone, so we flag this leakage risk explicitly and note that the developmental-stage mismatch (largely adult/bulk model tracks versus a fetal assay and eQTL anchor) works against, not for, inflated agreement. (ix) Broader engagement with prior reports of reporter–endogenous discordance (episomal versus integrated MPRAs, STARR-seq versus lentiMPRA, promoter-driven artefacts) and with endogenous perturbation and enhancer–gene-linking resources (CRISPRi/a screens; activity-by-contact and CRISPR-based maps such as Fulco *et al.* and Gasperini *et al.*) would further situate the endogenous benchmark.

4. Conclusion

Treating a DNA foundation model and an MPRA as two imperfect estimators, rather than model-versus-truth, converts a null correlation from a disappointing benchmark result into an informative one. Across 3,976 psychiatric-risk regulatory variants, AlphaGenome and Borzoi agree with each other but not with an integrated developing-cortex lentiMPRA; the disagreement is not measurement noise; the calibrated agreement is credibly zero; and an independent, fine-mapped, developmentally matched eQTL benchmark tracks the models, not the reporter. The honest deliverable is a calibrated concordance map of where sequence prediction and reporter measurement encode the same signal — and a recommendation that fine-mapped endogenous QTLs, not decontextualized reporters, are the appropriate arbiter of *in silico* regulatory variant-effect prediction. As a translational readout of the concordance map, resolving the direction of effect within the concordant core shows that magnitude concordance is only weakly predictive of sign agreement (10 of 25 concordant DAVs agree in direction across reporter and both models), and yields a ranked, orthogonally-supported shortlist for functional validation — headed by rs76980967→*MFGE8*, the single variant on which reporter, both foundation models, and a fine-mapped fetal-cortex eQTL all agree in direction, and thus the strongest candidate for CRISPR base editing at the native locus. Tracing the concordant core into Open Targets shows that every nominated gene is a novel, undrugged target, and identifies *STIP1* (a bipolar-disorder–associated, small-molecule-tractable gene with full direction agreement) as the strongest overall druggable candidate and *MFGE8* as the strongest by causal confidence — while making explicit that variant-to-indication assignment, not target tractability, is the rate-limiting step from a concordance map to a therapeutic hypothesis.

5. Methods

Variant set and reporter measurements

Variants and measured allelic effects were taken from the developing-human-cortex lentiMPRA of Deng et al. [1], covering psychiatric-risk regulatory variants (brain eQTLs/caQTLs and GWAS SNPs) in primary cortical cells and hiPSC-derived cerebral organoids. The lentiMPRA reporter is lentivirally delivered and stably integrated into the host genome, with the assay signal restricted to integrated (chromatinized) copies. We used the study's processed per-variant $\log_2$ fold change of alternate versus reference allele activity and its limma-moderated standard error and adjusted p-value. After QC (biallelic SNVs only; multi-allelic sites and indels excluded) the analysis set comprised 3,976 variants (hg38), of which 76 were differentially active (adjusted $p < 0.10$; DAVs). A reference-allele check against hg38 (UCSC, Ensembl fallback) confirmed allele orientation on sampled variants.

AlphaGenome and Borzoi variant scoring

Variants were scored with AlphaGenome (v0.7.0, API) in native genomic context; per-variant summary is the mean allelic difference (DIFF_LOG2) over brain/CNS tracks pooled across four scorers (DNase, ATAC, CAGE, ChIP-histone), 100 kb window (1 Mb tested as ablation, cross-context $\rho = 0.994$). Borzoi was run on GPU (524 kb windows) over brain DNase/CAGE/RNA tracks (reference-allele concordance 99.9%). Both summaries are robust to the specific aggregation ($\rho \geq 0.98$).

Independent fine-mapped eQTL benchmark

As an external endogenous anchor we used the developing-brain xQTL atlas [14] (github.com/gandallab/devBrain_xQTL, Supplementary Data release, mixed cis-eQTL SuSiE fine-mapping), which provides per-variant credible-set membership, posterior inclusion probability (PIP), z-score and posterior effect size. Atlas coordinates (hg19) were lifted to hg38 with the UCSC hg19→hg38 chain (122,016/122,050 variants lifted); 370 Deng variants matched an eQTL credible-set variant by exact chr:pos:ref:alt (0 required allele flip). Agreement was quantified with Spearman correlation of effect magnitude (95% CI by bootstrap, 2,000 resamples) and with signed direction-of-effect concordance stratified by PIP (binomial test vs 0.5). Partial Spearman correlations controlled for phyloP, baseline activity and reporter effect by rank-regression residualization.

Activity–Context–Propagation model

The ACP analysis models the SuSiE posterior eQTL effect Y_i (alt-oriented) from the oriented reporter effect M_i and native context, using posterior SD as known measurement error and PIP as a causal-confidence weight. Context features (cortical ATAC accessibility across cell types, activity-by-contact enhancer–promoter contact breadth and target-gene count, phyloP, motifbreakR disruption, reporter baseline activity, and a library-wide enhancer-redundancy count) were split by provenance — AlphaGenome/Borzoi tracks excluded from the primary axes — and reduced within groups by PCA to orthogonal composites (accessibility, coupling, conservation) plus a redundancy axis, combined into one standardized context-favorability score approximately orthogonal to PIP ($r = 0.06$) and reporter magnitude ($r = 0.03$). A frequentist nested ladder (null; reporter; context; additive; reporter×context; sequence-model; combined) was compared by repeated (5×40) PIP-weighted k-fold cross-validation with ridge regression, reporting weighted held-out R^2 and paired win fractions. The Bayesian model, written from scratch in NumPyro, is $Y_i \sim \mathrm{Normal}(\mu_i, \sigma_i)$ with $\mu_i = \beta_0 + \beta_M M_i + \Sigma_k \beta_k C_{ik} + \gamma(M_i \times fav_i)$, a regularized-horseshoe prior on context coefficients and γ, a PIP-tempered likelihood ($\sigma_{meas,i} = \mathrm{posterior\_SD}_i/\sqrt{\mathrm{PIP}_i}$), and an intrinsic-scatter term ($\sigma_i^2 = \sigma_{meas,i}^2 + \sigma_{extra}^2$, $\sigma_{extra} \sim \mathrm{HalfNormal}(0.1)$); inference used NUTS (4 chains, 1,500 warmup / 2,000 draws, target acceptance 0.95), with convergence by R-hat and effective sample size and fit by the standardized-residual distribution. Variants were classified into a 2×2 of reporter status (DAV call or top-tertile magnitude with FDR < 0.5) by endogenous status, the endogenous null defined from the posterior as $P(|Y| < \delta \mid \text{data}) > 0.8$ at $\delta = 0.05$. Context enrichment of the latent-potential class was tested against reporter-matched controls (nearest-neighbor matched on reporter magnitude, reporter uncertainty, PIP, activity and conservation) and by logistic regression of class membership on the context axes within the reporter-active pool.

Context-window ladder

For the context-type comparison (Section 2.10), AlphaGenome was additionally scored with a 1 Mb window (brain/CNS track mean, DIFF_LOG2) alongside the canonical 100 kb scoring. Agreement of each estimator's effect magnitude with the fine-mapped eQTL posterior-effect magnitude was computed by Spearman correlation with 95% bootstrap confidence intervals (2,000 resamples) on the 369 credible-set variants carrying a 1 Mb score; the estimator-proximity matrix is the pairwise Spearman ρ of the same magnitudes among the lentiMPRA, the two AlphaGenome windows and Borzoi. Cross-window agreement of the two AlphaGenome scorings was ρ = 0.93 on this credible-set subset and ρ = 0.99 genome-wide (all 3,976 variants).

Bayesian measurement-error model

A from-scratch hierarchical measurement-error model (NumPyro): measured effect $y_i \sim \mathrm{Normal}(\theta_i, s_i)$ with the limma standard error s_i as known noise and a regularized horseshoe prior [9] on θ_i, fit non-centered by NUTS over all 3,976 variants (max $\hat{R}$ = 1.007, no divergences). Posterior effects and inverse-variance weights were used to recompute agreement on denoised/reliability-weighted effects.

Latent-variable generative model and simulation-based calibration

A from-scratch latent-variable model (NumPyro): each variant has two signed latents (E, A) drawn from a bivariate normal with unit variances and correlation ρ; AlphaGenome, Borzoi and motifbreakR load on E, the lentiMPRA (heteroscedastic, using its limma SE) on A. All sources z-scored so each loading equals its correlation with its latent; ρ is identified through the cross-source covariance. Per-variant latents are marginalized analytically (linear-Gaussian core; Woodbury/matrix-determinant-lemma low-rank likelihood validated to 10^{-6} against a full multivariate normal), so NUTS samples only ~11 global parameters. Chains were run as independent single-chain fits (distinct seeds) and stacked for cross-chain $\hat{R}$ (2 chains × 800 draws). phyloP was held out as orthogonal validation. Simulation-based calibration: data were simulated from the fitted model at a grid of known ρ under the real heteroscedastic reporter noise and canonical loadings, then refit with the same model; we report HDI coverage and bias of the recovered ρ across the grid.

Additional sequence model (Enformer)

To reduce dependence on the AlphaGenome/Borzoi family, we scored all 3,976 variants with Enformer (public "official-rough" weights, 197 kb receptive field), computing alternate-minus-reference predictions over central bins of brain CAGE and DNase tracks (single L40S GPU, 18 min; all 3,976 variants scored). Enformer is a third, independently developed sequence model with a distinct training regime; its inclusion tests whether the estimand mismatch is a property of sequence-to-function models in general rather than of two related architectures. Results are reported in Section 2.14 and Figure 16.

Concordant-core case study and prioritization

The 25 concordant differentially active variants were taken as the intersection of the calibrated concordant class with the differential-activity call. For each we recorded the signed allelic effect of the lentiMPRA ($\log_2$ fold change and its horseshoe-shrunk posterior mean), AlphaGenome (canonical brain summary), Borzoi (brain-track signed mean), and, where available, the fine-mapped eQTL posterior effect; all were oriented to the alternate allele. Direction agreement was scored across the reporter and the two foundation models (all-agree, two-of-three, or models-oppose-reporter), and a fine-mapped eQTL direction was counted only for variants in a credible set with PIP ≥ 0.1 and a non-trivial posterior effect. The convergent-evidence score is the sum of seven bounded, equally-scaled axes — direction agreement, reporter effect magnitude and significance, fine-mapped eQTL support (credible-set membership, PIP tier, and signed agreement), psychiatric-GWAS linkage-disequilibrium count across nine disorders, motifbreakR TF-motif disruption, AlphaGenome brain accessibility and reporter baseline activity, and phyloP conservation — each capped so no single axis dominates. Predicted target genes were assigned from activity-by-contact enhancer–gene links (per cortical cell type), fine-mapped eQTL target, and motifbreakR-disrupted TF, in that priority. The matched comparison panel comprised the 8 top-ranked concordant variants, the 4 model-only (endogenous-latent-active/reporter-quiet) variants with the largest AlphaGenome effect, the 4 reporter-only variants with the largest reporter effect, and the 4 lowest-effect both-quiet variants, drawn from the same calibrated classes and compared on baseline reporter activity.

Target tractability and disease provenance

For the genes nominated by the concordant core we queried the Open Targets Platform (GraphQL API v4) for target tractability (small-molecule, antibody, and PROTAC/degrader modality buckets), genetic constraint (gnomAD loss-of-function observed/expected upper bound, LOEUF), reported safety liabilities, and disease associations. Each modality was assigned a tier — clinical/structural evidence, medium-confidence annotation, or predicted only — and genes were ranked by a composite of causal-pointer strength (gene-link tier plus direction agreement), best modality tier, and maximum psychiatric/neurological disease-association score. Disease provenance of each variant was established from its Open Targets credible sets: a variant was assigned a genome-wide psychiatric indication if in linkage disequilibrium with a genome-wide-significant psychiatric GWAS signal in the reference panel, and otherwise classified by the strongest available credible set (sub-threshold psychiatric GWAS, brain molecular QTL only, non-psychiatric GWAS trait, or none). LOEUF, tractability tiers, and disease provenance are provided as a per-gene table.

Orthogonal evidence and statistics

Independent estimators: motifbreakR PWM motif-disruption scores [5], phyloP conservation [6], and the fine-mapped eQTL atlas [14]. Agreement by Spearman rank correlation (denoted ρ throughout; *p* denotes a p-value, never a correlation); class separation by Mann–Whitney U and AUROC; a subgroup scan over candidate strata was assessed against a permutation null. Analyses used Python (NumPy, pandas, SciPy) with NumPyro/JAX [8] and ArviZ. Coordinate liftover used pyliftover with the UCSC chain; SuSiE fine-mapping is as provided by the atlas [20].

Data and code availability

Processed benchmark tables, model scores, the eQTL-overlap table, SBC results and all analysis scripts are provided as project artifacts with a pinned environment specification and fixed seeds. The full per-variant concordance map for all 3,976 variants — coordinates, each estimator's effect, and the calibrated posterior class probabilities ($P_{concordant}$, $P_{reporter\text{-}only}$, $P_{model\text{-}only}$, $P_{both\text{-}quiet}$) alongside the hard class call — is released as a documented supplementary table so downstream users can filter on posterior class probability rather than hard calls. Reporter measurements derive from Deng et al. [1]; the eQTL atlas from [14].

6. Reproducibility and Ethics

Reproducibility. This is a secondary reanalysis of published data; no new human or animal data were generated. Reporter measurements are from Deng et al. [1]; the fine-mapped eQTL benchmark is the developing-brain xQTL atlas [14] (SuSiE credible sets). AlphaGenome was accessed via its API (v0.7.0); Borzoi and Enformer were run from public pretrained weights (Enformer "official-rough", scored on one GPU). All variant coordinates are hg38; the eQTL atlas (hg19) was lifted with the UCSC chain. Every analysis script was written from scratch for this study and is provided as project artifacts with a pinned environment; random seeds are fixed and reported, and MCMC diagnostics ($\hat{R}$, divergences) are reported for every fit.

Ethics and data use. No identifiable individual-level genotype or phenotype data were accessed; all inputs are aggregate/summary-level published resources. The AlphaGenome API terms govern redistribution of its outputs; we release derived per-variant effect summaries needed to reproduce the figures, not raw API responses. The variant library is ascertained for psychiatric-disorder-associated regulatory variants; we make no clinical or diagnostic claims about any variant, and the per-variant concordance map is a research triage, not a basis for clinical interpretation.

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Supplementary Figures

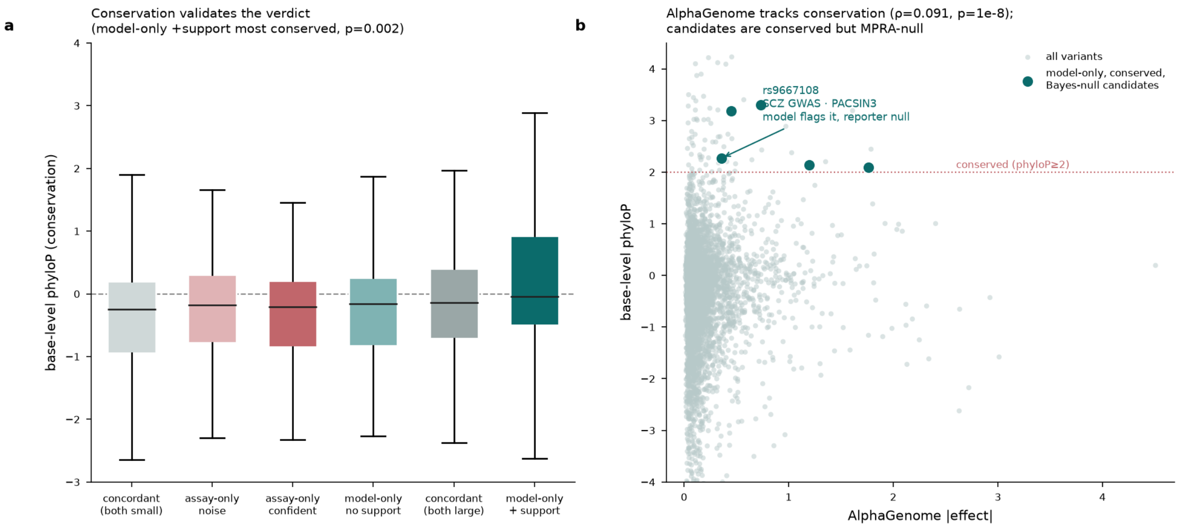

Figure S1. Base-level conservation discriminates model-supported divergent variants. (a) phyloP conservation by attribution class; model-supported variants are the most conserved (Mann–Whitney p = 0.002). (b) AlphaGenome effect magnitude versus base-level phyloP. *Panel labels retain earlier "verdict" terminology; per the assay-identity correction these classes are read as native-context-dependent vs locally-encoded.*

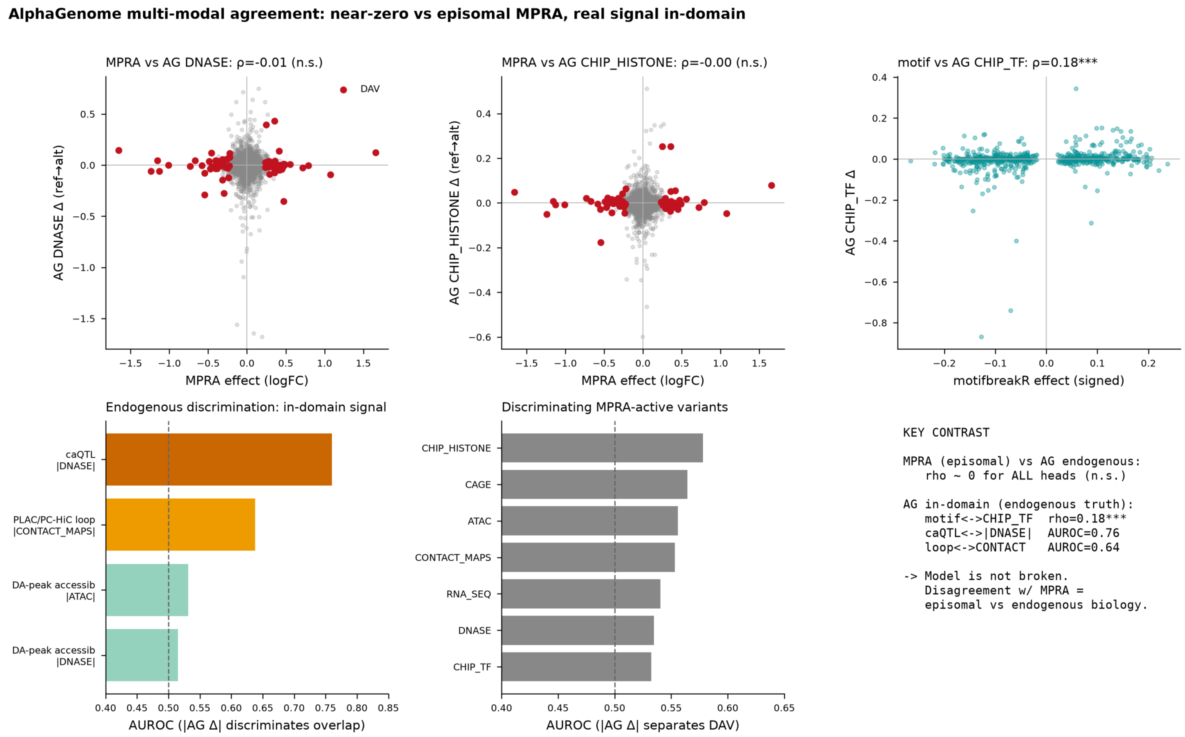

Figure S2. Multi-modal head sweep. AlphaGenome's seven output heads scored against every measured modality: no head correlates with the lentiMPRA effect, but the ChIP-TF head tracks signed motifbreakR disruption (ρ = 0.18) and the accessibility/contact heads discriminate caQTL overlap (AUROC 0.76) and chromatin loops (AUROC 0.64). *"Episomal" in panel text predates the assay-identity correction and should be read as "integrated reporter."*

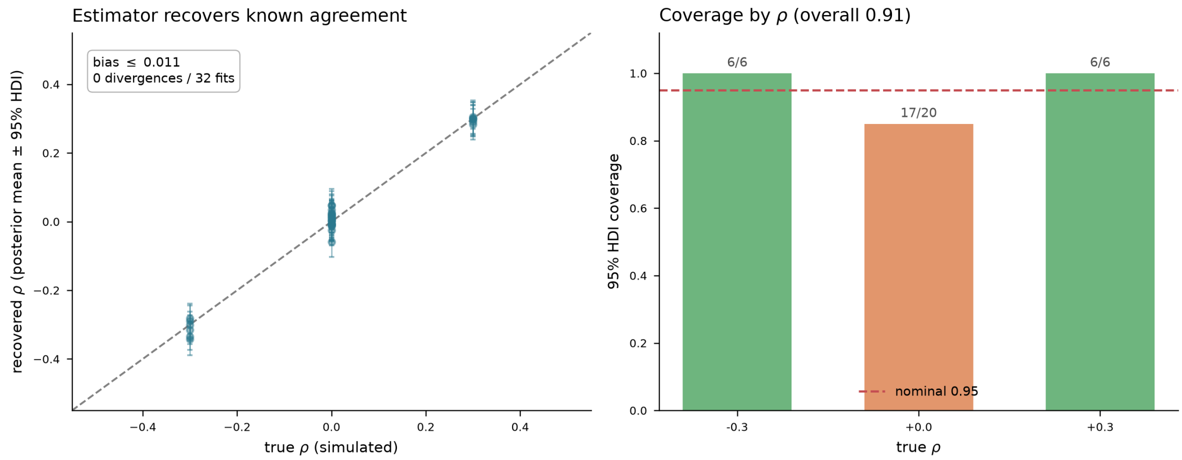

Figure S3. Simulation-based calibration of the latent-variable ρ estimator. Data simulated from the fitted generative model at known ρ (heteroscedastic reporter noise, canonical loadings) and refit; the estimator recovers true ρ with small bias (≤ 0.011) and 0 divergences over 32 fits. 95%-HDI coverage is shown per ρ with fit counts (6/6, 17/20, 6/6; overall 0.91); the ρ = 0 operating point (orange) is replicated 20× and is within binomial sampling error of the nominal 0.95.